

No trust in the choice architect? No problem! On the minor role of trust for the effectiveness of default interventions promoting the choice of energy-efficient appliances

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ABSTRACT

Pre-selected options, also known as defaults, have been shown to influence choices about everything from organ donations to retirement savings and mundane consumer decisions. Yet, boundary conditions of default effects need to be understood to determine their legitimacy as a behavioral policy instrument. Although default effects are multiply determined, they partly work through being perceived as recommendations by the source, the choice architect. This implies that trust in the choice architect may play a role in default effects. In default experiments in two countries – the Philippines ($n = 909$) and the U.S. ($n = 925$) – we manipulate trust. As expected, we find a strong default effect on the choice of energy-efficient appliances, which is of similar size across the two countries. The trust manipulation significantly influenced trust in the choice architect (the real estate agent) and the number of energy-efficient appliances chosen in the U.S. However, it did not change the default effect, which seems robust against mistrust in the choice architect. All in all, the study speaks for the robustness of default effects. The fact that the default effect is robust even when used by untrustworthy choice architects calls for regulation of its use.

In the endeavor of changing behavior to avoid a climate catastrophe, insights are needed on how and under what conditions policy interventions work as intended and when they do not. It is well documented that the effectiveness of behavioral interventions varies greatly (Bryan et al., 2021; Mertens et al., 2022; Milkman et al., 2021), and their general usefulness has been questioned (Benartzi et al., 2017; Grüne-Yanoff, 2012; McCrudden & King, 2015; Szaszi et al., 2018). Some argue that behavioral interventions can be useful and beneficial (Mertens et al., 2022), while others warn against misuse and the potential danger of deflecting attention and support from systemic government regulations (Chater & Loewenstein, 2022). Hence, it is essential to identify mechanisms and boundary conditions to determine the legitimacy of behavioral interventions as a policy instrument (de Ridder et al., 2022; Szaszi et al., 2018).

As a prime example, the use of pre-selected choice options, also known as defaults, which become effective if the actor does not actively choose something else, is one of the most popular and consistently

effective behavioral interventions to help people make beneficial decisions for society and themselves (Jachimowicz et al., 2019). In general, people tend to stick with the pre-selected option, which has been shown to influence choices in different domains such as organ donations (Abadie & Gay, 2006; Johnson & Goldstein, 2003), medical decisions (Ansher et al., 2014; Halpern et al., 2013), retirement savings (Benartzi et al., 2017; Choi et al., 2003, 2004; Madrian & Shea, 2001), and organic grocery shopping (Kuhn et al., 2022). In relation to pro-environmental behavior, defaults have been most prominently used to motivate energy savings by defaulting households to sign up for “green electricity” tariffs (Allcott & Mullainathan, 2010; Ebeling & Lotz, 2015; Pichert & Katsikopoulos, 2008; Vetter & Kutzner, 2016), to choose energy-efficient appliances (Steffel et al., 2016; Van Gestel et al., 2021), to make use of daylight instead of electric lighting (Heydarian et al., 2016), or to accept “smart” regulation of their electricity consumption (Toft et al., 2014). However, the boundary conditions for the effectiveness of defaults have received relatively little attention.

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One factor that has been suggested to influence the effectiveness of behavioral interventions is trust in the source, the choice architect (Krijnen et al., 2017). A recent meta-analysis shows that trust strongly predicts climate-friendly behaviors (Cologna & Siegrist, 2020). Behavioral policies brought forward by trusted sources are more likely to be accepted (Evers et al., 2018; Junghans et al., 2015). Beyond studies on policy acceptance, trust as a potential moderator of the effectiveness of behavioral interventions has not been tested. This pre-registered study investigates trust as a boundary condition of default intervention effects by experimentally manipulating trust in the choice architect.

Beyond the specific trust in the choice architect, general trust in people, which varies by country (Falk et al., 2018), could potentially impact the effectiveness of a default intervention. Most studies on the effectiveness of behavioral interventions have been conducted in homogenous samples drawn from Western, educated, industrialized, rich, and democratic (WEIRD) countries (Henrich et al., 2010). A recent review points out that environmental psychological research lacks studies with cross-cultural samples (Tam & Milfont, 2020). The current research seeks to close the knowledge gap on some crucial, possible boundary conditions of default effects by using the choice of energy-efficient, electric household appliances in a WEIRD and a non-WEIRD country as the case.

Specifically, we investigate default effects in the context of the choice of household appliances in the U.S. and the Philippines. For energy-efficiency behavior, a comparison between these two countries is interesting for several reasons. The countries vary in general trust, with the Philippines scoring higher than the U.S. (Falk et al., 2018). Regarding household energy consumption, both countries vary greatly as well. The U.S. has one of the highest per capita energy consumption levels worldwide, far more than the Philippines. Still, in both countries, the largest share of domestic electricity consumption is due to refrigeration and air conditioning (RAC). Whereas the U.S. has the world's highest concentration of air conditioners (nearly 90% of households used air conditioning in 2020 (EIA, 2020)), the Philippines is the country with the most significant increase in cooling needs (Atalla et al., 2018) and experience an accelerating demand for RAC appliances (GIZ, 2019). From a behavioral perspective, purchase decisions for RAC devices profoundly impact a household's carbon emissions. The choice task in this study tests preferences regarding the energy-efficiency of RAC appliances and two other high-consumption electronic devices. To allow for experimental manipulation of the trust in the choice architect, we adopt a popular default choice task introduced by Steffell et al. (2016).

1. Boundary conditions and theoretical explanations of default effects

A recent meta-analysis ascribes considerable effectiveness to defaults ($d = 0.68$, 95% confidence interval = 0.53–0.83), but at the same time, finds great variation, depending on whether specific psychosocial channels are triggered (Jachimowicz et al., 2019). At the same time, research on the boundary conditions of default effects has produced contradictory evidence (de Ridder et al., 2022). Some have argued that default interventions preserve freedom of choice and should be moderated by people's attitudes (Johnson & Goldstein, 2003; Thaler & Sunstein, 2008). Previous studies testing this assumption have revealed that neither disclosing the presence and/or purpose of a default (Bang et al., 2018; Bruns et al., 2018; Loewenstein et al., 2015; Steffell et al., 2016) nor people's preferences (Bronchetti et al., 2013; Kuhn et al., 2021; Steffell et al., 2016; Taube & Vetter, 2019; Theotokis & Manganari, 2015; Van Gestel et al., 2021; Vetter & Kutzner, 2016) change the effectiveness of defaults. Still, it is too early to conclude that the default effect has no boundaries. Rather, the boundary conditions are still poorly understood and need further research.

1.1. Physical effort

One theoretical explanation for default effects is their potential to

free the decision-maker from cognitive elaboration and physical effort by passively staying with the (supposedly beneficial) default option (Samuelson & Zeckhauser, 1988; Thaler & Sunstein, 2008). For example, it was found that a supposedly tiresome series of product customization decisions involving a multitude of options increased adherence to the default (Levav et al., 2010). However, no moderation was found in a recent study in which de-selecting the default was made more effortful (Kuhn et al., 2021). Hence, so far, the evidence for moderation by effort is not conclusive.

1.2. Loss-aversion

The second theoretical explanation holds that defaults change the psychological representation of choice options. The pre-selected option might be evaluated as an endowment (Park et al., 2000), and forgoing it might be perceived as a loss (Tversky & Kahneman, 1991). It has been confirmed that defaults change the perception of the choice options and spontaneously attract more positive thoughts toward the default option (Dinner et al., 2011). Changing the decision situation by prompting the default option with additional information that increases attention toward the default option also increases its effect on choices (Betz et al., 2022). Still, studies in which attention was drawn toward the presence of the default by disclosing it did not find a moderation effect (Bang et al., 2018; Bruns et al., 2018; Loewenstein et al., 2015; Steffell et al., 2016). These findings speak against this theoretical explanation.

1.3. Implied endorsement

A third theoretical explanation suggests that defaults are taken to be a recommendation. Research has shown that defaults can be perceived as a recommendation by others (Everett et al., 2015) and by the choice architect (McKenzie et al., 2006). If it is assumed that defaults are deliberately set by the choice architect – as a recommendation of the option that people ought to select or as information about which option, the majority would select – this implies that the default signals the choice architect's attitude and wishes for the behavior of others. For example, policymakers usually select their country's "organ donor" default. It seems reasonable to assume that the policymakers believe this to be the most ethical choice option for the population (Johnson & Goldstein, 2003). It has been found that decision-makers develop inferences about the preferences of choice architects, that organ-donation defaults are perceived as recommendations by policymakers (McKenzie et al., 2006) and that product-configuration defaults are perceived as recommendations by retailers (Brown & Krishna, 2004). In line with this theoretical explanation, trusting that the choice architect has the best interests of the target group in mind could play a crucial role in the effectiveness of defaults.

To our knowledge, no studies have examined how the perception of the choice architect influences the effectiveness of defaults. On the backcloth of the ongoing debate on the ethicality of behavioral interventions (Feitsma, 2018; Lepenies & Malecka, 2018; Schubert, 2017), it is particularly problematic if defaults interventions can be effectively used even by untrustworthy choice architects.

2. Trust as a potential moderator of default effects

The success of behavioral policy interventions depends on the target population's assessment of the beliefs and intentions of the choice architect (Krijnen et al., 2017). It especially depends on the extent to which decision-makers trust that the choice architect aims to uphold and promote their best interests and values with the behavioral intervention. Decision-makers who believe that the choice architect holds malevolent intentions should be less likely to accept the pre-selection and choose the default option. Brown and Krishna (2004), examining implied endorsement as a mechanism, found that participants were more likely to create counterarguments and to de-select an expensive default option

when they received information about the financial motivation of the retailer as choice architect. Being faced with a choice situation under the control of a mistrusted choice architect can also create reactance (Brehm, 1966), possibly prompting an active de-selection of the default option. In turn, higher trust should also lead to a stronger default effect. Advice from trusted sources positively influences decisions (Kinney et al., 1998). In one experiment, the default was more effective if a trustworthy choice architect, an administration of a public university, was made transparent (Paunov et al., 2018 Study 1 & 2). Hence, it is likely that trust in the choice architect moderates the default effect, in line with the theoretical explanation of implied endorsement.

In sum, most previous research on the perception of the choice architect has studied its relation to the acceptance rather than the effectiveness of behavior change interventions (de Ridder et al., 2022). Approval of behavioral policy has been shown to increase with the expertise and trustworthiness of the source (Evers et al., 2018; Junghans et al., 2015). It has also been suggested that the acceptance of behavioral policies decreases if the source is mistrusted (Jung & Mellers, 2016). For example, people find the general use of behavioral interventions more acceptable when suggested by their associated, trusted political party. However, they view those same interventions as more unethical when suggested by another political party (Tannenbaum et al., 2017). Yet, to our knowledge, studies that directly examine the moderating impact of trust on the effectiveness of default interventions are still missing.

3. Predicting energy-efficient choices

A recent meta-analysis found that climate-friendly behaviors depend on trust (Cologna & Siegrist, 2020). For example, trust in other people predicts sustainable purchase decisions (Gupta & Ogden, 2009), daily energy-saving behaviors (Fuhrmann et al., 2021), and also the willingness to invest in energy cooperatives (Fischer et al., 2021) or green energy contracts (Ziegler, 2021). A lack of trust or explicit mistrust has been found to be a barrier to pro-environmental behaviors (Nuttavuthisit & Thøgersen, 2017). Trust is also an essential precondition for the effectiveness of product sustainability labeling that, for example, determine purchase decisions for energy-intensive goods (Daugbjerg et al., 2014; Issock et al., 2018; Schuitema et al., 2020). It, thus, can be assumed that general trust in other people increases energy-efficient choices, while specific trust in the choice architect should increase the effect of behavioral policy interventions.

Variations in general trust in other people could also explain country differences in supporting behavioral policies such as energy-efficiency measures. Populations in countries with little confidence in their government are generally less supportive of behavioral interventions (Sunstein et al., 2018). According to the Global Preference Survey (Falk et al., 2018) and the World Values Survey (Haerpfer et al., 2022), using different measures, South-East Asian countries, such as the Philippines, are generally less trusting than WEIRD countries, such as the U.S. or Western European countries.

Many other factors that might impact energy-efficient choices also vary across cultures and countries, including, for example, risk tolerance and time preferences (Falk et al., 2018). Research shows that more risk-averse consumers are less likely to adopt energy-efficient technologies such as efficient domestic appliances (Farsi, 2010; Fischbacher et al., 2021; Qiu et al., 2014). Risk perceptions also meaningfully differ by country (Siegrist & Árvai, 2020). For example, people in South-East Asia appear slightly less willing to take risks than in North America (Falk et al., 2018). Additionally, more patient people, who discount future outcomes less, are more likely to invest in energy-efficient technologies (Figuerola et al., 2019; Fuerst & Singh, 2018; Joshi & Rahman, 2015; Qiu et al., 2014) and consume less energy (Fuhrmann et al., 2021). More future-oriented individuals tend to live in homes with higher energy-efficiency and lower energy costs (Fischbacher et al., 2021). People in high-income countries appear, on average, to be more patient than those in low-income countries (Falk et al., 2018; Ruggeri

et al., 2022).

Prior research in the environmental domain has generally found that default effects do not depend on pro-environmental attitudes (Bronchetti et al., 2013; Steffel et al., 2016; Taube & Vetter, 2019; Theotokis & Manganari, 2015; Van Gestel et al., 2021). At the same time, there is strong evidence that environmental concern positively influences pro-environmental behavior, independently of defaults (Kuhn et al., 2021; Taube & Vetter, 2019). For example, the choice of energy-efficient appliances increases with environmental concern in WEIRD countries such as Canada and the U.S. (Berman Caggiano et al., 2021), as well as in non-WEIRD countries such as Ghana and the Philippines (Kuhn et al., 2022).

In sum, based on the reviewed literature, we predict that consumers are more likely to choose energy-efficient appliances when set as default and when the trust in the choice architect is higher, and, critically, that higher trust in the choice architect will lead to a stronger default effect (see Fig. 1). Furthermore, we predict that environmental concern, risk tolerance, patience, and general trust lead to stronger preferences for energy-efficient appliances.

4. Method

4.1. Participants and design

The study was pre-registered on [AsPredicted.org](https://aspredicted.org/yn3if.pdf) (<https://aspredicted.org/yn3if.pdf>). Participants were sampled from an online panel provided by Qualtrics. The costs of using a Qualtrics panel are significantly higher than other online panels (e.g., Amazon Mechanical Turk or Turk Prime). However, Qualtrics panels are known for better data quality (Sharpe Wessling et al., 2017). The target population was middle class or higher, living in a suburban or urban area, and at least 18 years old. To be eligible, participants had to own an air conditioner, a refrigerator, a freezer, an electric oven and stove, and a washing machine. Based on prior research, we considered a small effect size of interest. Hence, we conducted a power analysis with the statistical software R, with the "pwr.f2.test()" function of the "pwr" package, with a minimum effect size of interest of $f^2 = 0.02$ (small), $\alpha = 0.05$, and power = 0.80. For a model with three predictors (model 3), the power analysis suggested a minimum sample size of 549. For a model with seven predictors (model 4), the analysis suggested a minimum sample size of 724. To be on the safe side, we collected a final sample of 909 participants in the Philippines (543 females, 359 males, seven genderqueer/nonbinary) and 945 participants in the U.S. (562 females, 383 males, 0 genderqueer/nonbinary). These numbers are after excluding 76 speeders in the Philippines and 87 speeders in the U.S. (participants who took less than half of the median time). With these samples, still assuming a power of 80% with seven predictors, the smallest detectable effect size is $f^2 = 0.016$ for the Philippines and $f^2 = 0.015$ for the U.S. For the Philippine sample, the average age is 32.5 years ($SD = 9.96$), 99.3% have a high school degree or higher, the majority (77.3%) have a bachelor's or master's degree, 83.4% are working, and 0.9% are retired. For the U.S. sample, the average age is 60.8 years ($SD = 15.17$), 99.7% have a high school degree or higher, the majority (54.3%) have a bachelor's or master's degree, 41.6% are working, and 45.5% are retired. Based on census data, we calculated the average age of the adult population (≥ 18 years old) to be 35.0 in the Philippines and 47.3 in the U.S. in 2020–21. The difference in average age between the two samples is further amplified by the sampling basis being restricted to residents of urban and suburban areas owning a range of electrical appliances. In the Philippines, these restrictions made the sample slightly younger, and in the U.S., somewhat older. Therefore, we control for age in the statistical analyses. However, the age composition is part of the primary national characteristics of the two countries and, therefore, potentially one of the reasons for differences between countries.

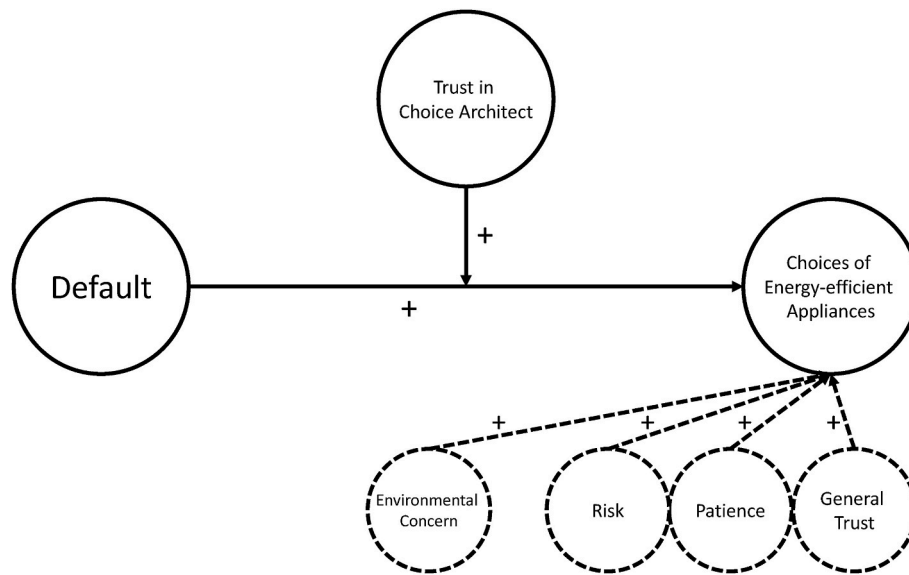


Fig. 1. Conceptual model tested in this paper, where default is the primary predictor. The influence of the default on the number of energy-efficient appliances is moderated by trust in the choice architect. Environmental concern, risk tolerance, patience and general trust are included as covariates. Experimental manipulated variables are depicted with solid lines, Correlational variables are depicted with dashed lines.

4.2. Design & procedure

To measure the effectiveness of defaults and trust in promoting energy-efficient choices, we used a scenario where participants had to imagine that they were about to move to a newly constructed apartment they had rented. For the apartment, final choices on the to-be-installed appliances (standard or energy-efficient) had to be made. The task was adapted from Steffel et al. (2016; also see Van Gestel et al., 2021). We manipulated trust with newspaper headlines. The headlines provided background information on local real estate companies, including the company responsible for their apartment. The headlines were either formulated to enhance trust or mistrust towards real estate. We used a 2 (energy-efficient vs. standard default) $\times$ 2 (trust vs. mistrust) between-subjects design with the number of energy-efficient appliances chosen as the dependent variable.

After participants had provided their informed consent, we checked their eligibility via socio-demographic background questions. Background questions included age, class, location, and appliance ownership (see above for eligibility criteria). Participants who were not eligible received a message accordingly and were thanked for their interest. We then administered three questions from the Global Preference Survey on patience, risk tolerance, and general trust (Falk et al., 2018 see below for a description of each measure). To reduce possible instrument effects, we measured those before the experimental manipulations. Next, participants were introduced to the scenario and randomly allocated to the trust or mistrust condition. After the trust manipulation, we checked participants' levels of attention and used a manipulation check to assess the effectiveness of the trust manipulation. Inattentive participants were not allowed to proceed. They received a message accordingly and were thanked for their interest. Attentive participants were then introduced to the appliance choice task. The task entailed the default manipulation based on Steffel et al. (2016, see below for more details). Adjoining the task, we measured environmental concern (modeled after Thøgersen et al., 2010).¹ We assume that environmental concern is less likely to be affected by the experimental manipulation, but might be more likely to

affect the decision-making process and therefore measured it after the task. Finally, we asked participants to rate the plausibility of the scenario and choice task, asked for additional demographics, inquired about suspicions concerning the goal of the study, monitored the amount of participation in other studies in the last half year. At the end participants were debriefed and thanked.

4.3. Materials

Specific trust manipulation. The choice architect was introduced as the "Better Living Real Estate Group." Participants were told that they had to fill out a form by Better Living to choose appliances for their new home. They were confronted with five newspaper headlines before the choice task. Headlines were based on country-specific news and carefully chosen to be as similar as possible in the two countries (see Appendix). One of the headlines expressly referred to the real estate company renting and outfitting the apartment. The headlines either were formulated in a way to either enhance trust or mistrust towards real estate. Participants were told that the headlines provided background information regarding the real estate sector in their country. As a bogus task, they were asked to indicate on a 5-point Likert scale how interesting they perceived each headline. As a manipulation check, participants were asked to indicate their trust in the choice architect Better Living and their general trust in real estate companies on a 10-point Likert scale (see Appendix).

Attention check. We tested participants' degrees of attention to the headlines by asking them about the content of the headline related to Better Living (see Appendix). Only participants who correctly answered the question were allowed to proceed with the study; 321 participants in the Philippines and 437 in the U.S. failed the attention check. The survey automatically ended early for those participants. They were excluded from the final sample.

Appliance choice task. Participants then received a form through which they could indicate their choice of five appliances (see Appendix). Instead of the 14 appliances used by Steffel et al. (2016), we only included appliances that consume a lot of energy and are frequently owned in the U.S. and the Philippines. In the opt-in condition standard options for each appliance were pre-selected, whereas in the opt-out condition energy-efficient options were pre-selected. Participants were told that, compared to the standard appliances, the energy-efficient appliances were more expensive, but the operating costs were lower

¹ For exploratory purposes, we also measured participants' perceived freedom to choose, or perceptions of external influence on their choices, based on Shen and colleagues (Dillard & Shen, 2005; Shen & Dillard, 2005). The measure was not included in any analysis.

Table 1
Hierarchical regression analysis predicting the number of energy-efficient appliances chosen in the Philippines ($N = 909$) and in the U.S. ($N = 945$).

Constant	Philippines					U.S.				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
	0.00 \ 3.49***	0.00 \ 3.49***	0.00 \ 3.49***	0.00 \ 3.49***	0.00 \ 3.43***	0.00 \ 3.58***	0.00 \ 3.57***	0.00 \ 3.57***	0.00 \ 3.57***	0.00 \ 3.69***
<i>Independent variables:</i>										
Default condition ($-0.5 = \text{standard}$, $0.5 = \text{efficient}$)	0.52 \ 2.08***	0.52 \ 2.08***	0.52 \ 2.09***	0.52 \ 2.10***	0.52 \ 2.09***	0.51 \ 2.08***	0.52 \ 2.09***	0.52 \ 2.09***	0.52 \ 2.11***	0.51 \ 2.08***
Trust condition ($-0.5 = \text{mistrust}$, $0.5 = \text{trust}$)		$-0.01 \ -0.04$	$-0.01 \ -0.04$	$-0.01 \ -0.06$	$-0.02 \ -0.07$		0.14 \ 0.57***	0.14 \ 0.57***	0.13 \ 0.51***	0.13 \ 0.51***
Default X trust condition		$-0.01 \ -0.06$	$-0.01 \ -0.04$	$-0.00 \ -0.04$	$-0.01 \ -0.05$			$-0.05 \ -0.41$	$-0.06 \ -0.50$	$-0.06 \ -0.51$
<i>Covariates:</i>										
Environmental concern				0.07 \ 0.14*	0.06 \ 0.13*				0.23 \ 0.40***	0.23 \ 0.40***
Patience				0.02 \ 0.03	0.03 \ 0.04				$-0.04 \ -0.04$	$-0.04 \ -0.03$
Risk				0.02 \ 0.04	0.03 \ 0.04				$-0.03 \ -0.02$	$-0.03 \ -0.00$
General trust				$-0.00 \ -0.00$	0.00 \ 0.00				$-0.02 \ -0.02$	$-0.04 \ -0.03$
Age					0.05 \ 0.01				0.12 \ 0.02***	0.12 \ 0.02***
Gender ($-0.5 = \text{male}$, $0 = \text{nonbinary}$, $0.5 = \text{female}$)					0.05 \ 0.19				0.04 \ 0.18	0.04 \ 0.18
Income					0.01 \ 0.01				0.06 \ 0.09*	0.06 \ 0.09*
Area ($-0.5 = \text{suburban}$, $0.5 = \text{urban}$)					0.04 \ 0.18				$-0.01 \ -0.06$	$-0.01 \ -0.06$
Adjusted R ²	0.27	0.27	0.27	0.28	0.28	0.26	0.28	0.28	0.33	0.34

Note: Standardized coefficients are shown before and unstandardized coefficients after the backslash. See Appendix 9 for a more detailed table that includes confidence intervals. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

due to the savings on the electricity bill. In a table, we presented choice options in both conditions symmetrically, next to each other. We balanced the horizontal order of appliances and the vertical order of choice options within the table. The dependent variable is the number of energy-efficient appliances selected by participants, ranging from 0 to 5, where a higher value indicates the choice of more energy-efficient appliances.

Environmental concern. Environmental concern was measured with six items using a 7-point Likert scale based on Thøgersen et al. (2010; see Appendix). The scale was highly reliable for the Philippines ($\alpha = 0.88$, $M = 5.64$, $SD = 0.96$) as well as the US ($\alpha = 0.94$, $M = 4.84$, $SD = 1.18$). Hence, the construct was represented by the average of all items, mean-centered, and used as a covariate.

Global preferences. We used the self-assessment items from the global preference survey (Falk et al., 2018) to measure general trust, patience, and risk tolerance using an 11-point Likert scale (see Appendix). All variables were mean-centered and used as covariates.

Demographics. Participants were asked about their age, gender, highest education level, and employment status. Furthermore, we inquired about suspicions concerning the aim of the study with one open-ended question. Also, participants were asked to report the number of studies they had partaken during the last half year.

5. Results

All data analyses were conducted in R (R Core Team, 2021). We analyzed each country's data with linear effect models using the `lm()` command from base R. All regression analyses use mean-centered predictors for continuous variables and effect coding for categorical variables.

5.1. Trust manipulation check

To test whether the trust manipulation affected trust in the choice architect, Better Living, we ran independent *t*-tests for each country. In the Philippines, participants in the trust condition ($M = 9.15$, $SD = 1.55$), compared to participants in the mistrust condition ($M = 6.55$, $SD = 2.78$), indicated significantly higher trust in the choice architect, $t = 17.34$, $p < 0.001$. The same pattern arose in the U.S., where participants in the trust condition ($M = 7.60$, $SD = 2.17$), compared to participants in the mistrust condition ($M = 3.53$, $SD = 2.46$), also indicated significantly higher trust in the choice architect, $t = 26.91$, $p < 0.001$.

5.2. Main analyses

For each country, we conducted a hierarchical multiple regression analysis to predict the number of energy-efficient appliances chosen by participants, starting in model 1 with the default condition (with the standard default coded as -0.5 and the energy-efficient default coded as 0.5), adding the trust condition in model 2 (with mistrust coded as -0.5 and trust coded as 0.5), the interaction between the two in model 3, and the covariates in model 4 (Table 1). The analyses revealed a significant main effect for the default condition both in the Philippines ($B = 2.08$, $p < 0.001$) and in the U.S. ($B = 2.08$, $p < 0.001$). The analysis shows that when energy-efficient appliances were pre-selected (i.e., the default), participants, on average, chose at least two energy-efficient appliances more than when standard appliances were pre-selected. When the trust condition was added to the model, explained variance improved slightly, and a significant main effect appeared for the trust condition in the U.S. ($B = 0.57$, $p < 0.001$) but not in the Philippines ($B = -0.04$, $p > 0.1$). Hence, when U.S. participants, but not participants from the Philippines, had been exposed to mistrust-enhancing (compared to trust-enhancing) headlines, they chose significantly fewer energy-efficient appliances. The interaction between the default condition and trust is not significant in the Philippines but significant in the U.S. when controlling for the covariates. However, contrary to the hypothesis, the

Table 2
Hierarchical regression analysis predicting the number of energy-efficient appliances chosen with the two country datasets combined ($N = 1854$).

Constant	(1)	(2)	(3)	(4)	(5)	(6)
	−0.00 \ 3.53***	0.00 \ 3.53***	−0.00 \ 3.53***	−0.00 \ 3.52***	−0.00 \ 3.52***	−0.04 \ 3.52***
<i>Independent variables:</i>						
Country (−0.5 = Philippines, 0.5 = U.S.)	0.02 \ 0.09	0.02 \ 0.08	0.02 \ 0.08	0.07 \ 0.29**	−0.01 \ −0.06	−0.00 \ 0.01
Default condition (−0.5 = standard, 0.5 = efficient)	0.52 \ 2.08***	0.52 \ 2.09***	0.52 \ 2.09***	0.52 \ 2.10***	0.52 \ 2.09***	0.52 \ 2.08***
Default condition X country	−0.00 \ −0.01	0.00 \ 0.01	0.00 \ 0.01	0.00 \ 0.01	−0.00 \ −0.02	−0.00 \ −0.01
Trust condition (−0.5 = mistrust, 0.5 = trust)		0.07 \ 0.26***	0.07 \ 0.26**	0.05 \ 0.21**	0.05 \ 0.21**	0.06 \ 0.22**
Trust condition X country		0.08 \ 0.62***	0.08 \ 0.61***	0.08 \ 0.60***	0.08 \ 0.62***	0.07 \ 0.58***
Default X trust condition			−0.03 \ −0.23	−0.03 \ −0.25	−0.03 \ −0.26	−0.03 \ −0.28
Default X trust condition X country			−0.02 \ −0.35	−0.03 \ −0.48	−0.03 \ −0.48	−0.03 \ −0.46
<i>Covariates:</i>						
Environmental concern				0.17 \ 0.31***	0.17 \ 0.30***	0.15 \ 0.27***
Patience				−0.01 \ −0.01	−0.00 \ −0.00	−0.00 \ −0.00
Risk				−0.00 \ −0.00	0.01 \ 0.01	0.02 \ 0.02
General trust				−0.02 \ −0.01	−0.02 \ −0.02	−0.02 \ −0.02
Age					0.14 \ 0.01***	0.12 \ 0.01***
Gender (−0.5 = male, 0 = nonbinary, 0.5 = female)					0.05 \ 0.19*	0.05 \ 0.19*
Income					0.04 \ 0.05	0.04 \ 0.05*
Area (−0.5 = suburban, 0.5 = urban)					0.02 \ 0.07	0.01 \ 0.06
E. concern X country						0.08 \ 0.27***
Patience X country						−0.03 \ −0.07
Risk X country						−0.02 \ −0.03
General trust X country						−0.02 \ −0.04
Age X country						0.03 \ 0.01
Gender X country						−0.00 \ −0.01
Income X country						0.03 \ 0.08
Area X country						−0.03 \ −0.24
Adjusted R ²	0.27	0.28	0.28	0.30	0.31	0.31

Note: Standardized coefficients are shown before and unstandardized coefficients after the backslash. See [Appendix 10](#) for a more detailed table that includes confidence intervals. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

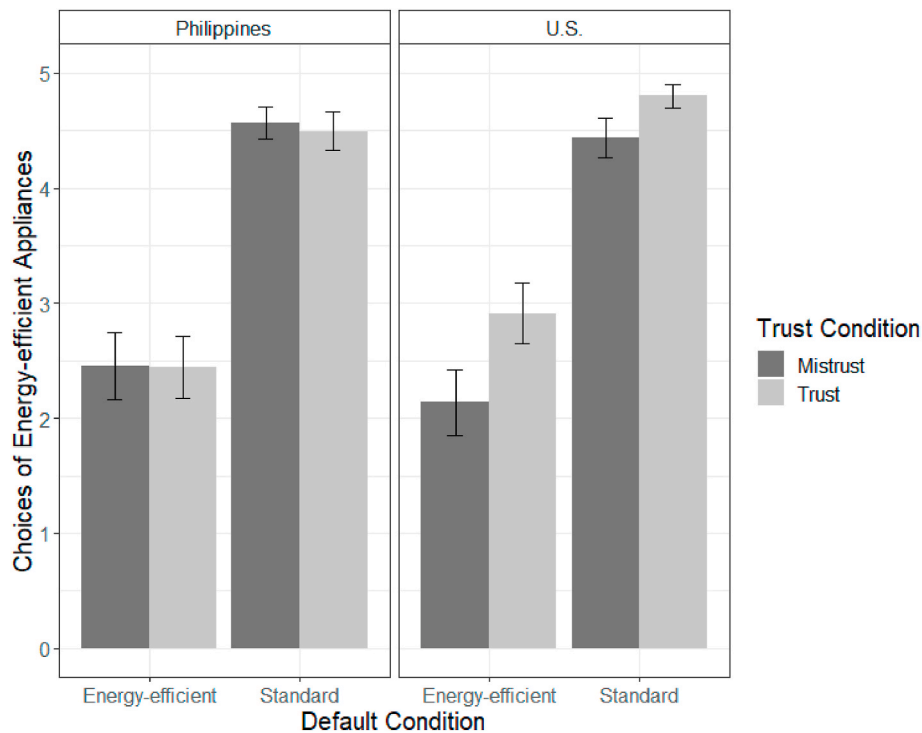


Fig. 2. Mean number of energy-efficient appliances chosen by experimental condition. Dots represent the raw data points.

effect is negative in the U.S. ($B = -0.50, p < 0.05$), which means that the default effect is bigger under mistrust than under trust conditions and that the effect of exposure to trust vs. mistrust-enhancing headlines is bigger when standard appliances are pre-selected than when energy-efficient appliances are pre-selected.

Furthermore, among the included covariates, only environmental concern significantly influenced the number of energy-efficient appliances being chosen, both by participants in the Philippines ($B = 0.14, p < 0.05$) and in the U.S. ($B = 0.40, p < 0.01$). As expected, the number of energy-efficient appliances chosen increased with participants' environmental concern. No other covariates were significant ($p > 0.21$).

5.3. Robustness check

In addition to our pre-registered analyses, we explored the robustness of the default effect by merging both datasets. We again used hierarchical multiple regression analysis to predict the number of energy-efficient appliances chosen, and we now included the country (coded as -0.5 for the Philippines and 0.5 for the U.S.) as a predictor. For each model, we included the main effect and the interaction with the country variable (Table 2).

The main effect of the country is significant when including covariates ($B = 0.30, p < 0.01$), as illustrated in Model 4. Participants in the U.S. choose more energy-efficient appliances than in the Philippines. Furthermore, the analyses found a significant main effect for the default condition ($B = 2.08, p < 0.001$), showing that participants tended to stick to the default option, choosing more energy-efficient (standard) appliances when they were pre-selected. The interaction between the country and the default condition is not significant, suggesting similar default effects in both countries. There is a significant main effect of the trust condition ($B = 0.26, p < 0.001$) and a significant interaction between the country and the trust condition ($B = 0.62, p < 0.001$), reflecting that the trust manipulation had an impact on choices in the U.S. but not in the Philippines. Importantly, according to this calculation, trust does not moderate the default effect ($B = -0.23, p = 0.14$), and this does not depend on the country ($B = -0.35, p = 0.27$). As we saw before, environmental concern positively influences the number of energy-

efficient appliances chosen ($B = 0.31, p < 0.001$). There is a positive interaction between environmental concern and country ($B = 0.26, p < 0.01$), reflecting that the impact of environmental concern on the choice of energy-efficient appliances is significantly stronger in the U.S. than in the Philippines. No other covariates are significant ($p > 0.11$).

In sum, in both countries when energy-efficient appliances were pre-selected (i.e., the default option) participants chose more energy-efficient appliances than when the standard appliances were pre-selected and they did so to a similar degree. This speaks to the robustness of the default effect across a WEIRD and a non-WEIRD country context. Situationally manipulated trust in the choice architect affected choices only in the U.S., where participants in the trust condition chose more energy-efficient appliances. However, contrary to our hypothesis, we did not find that situationally induced mistrust (vs. trust) decreased the default effect in any of the countries. In the U.S., the default effect appeared slightly higher for participants in the low trust condition. The trust manipulation appeared to impact the choice of energy-efficient appliances less when pre-selected than when standard appliances were pre-selected (Fig. 2).

6. Discussion

This study adds to the evidence for the robustness of default effects. Across a non-WEIRD and a WEIRD country, the chosen default had an identical, significant, and substantial impact on the choice of energy-efficient appliances. In both the Philippines and the U.S., participants were more likely to choose the pre-selected option and choose more energy-efficient appliances when they were the default.

Importantly, the default effect was not influenced by the situational manipulation of the trustworthiness of the choice architect. The trust manipulation successfully influenced participants' trust in the choice architect. Yet, it did not seem to moderate the default effect (Table 2). Only in the U.S. did mistrust lower energy efficient choices irrespective of the default. There are several possible reasons for this. First, participants might not necessarily have perceived the energy-efficiency default as (only) the endorsement by the immediate choice architect (i.e., the "Better Living Real Estate Group") but also by a wider range of societal

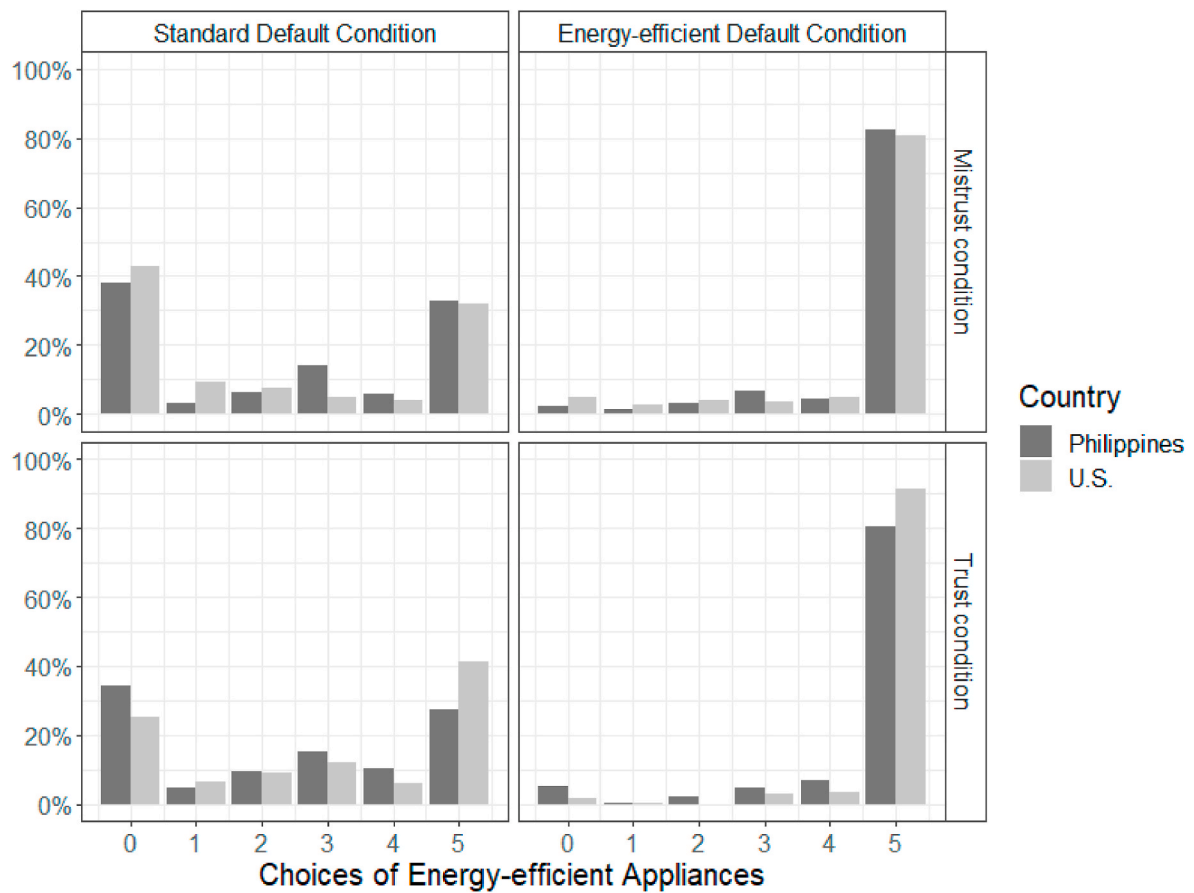


Fig. 3. Percentage of choices of energy-efficient appliances by each experimental group.

stakeholders, which might have reduced the negative effect of inducing mistrust in the real estate agent. Second, recall that participants were asked to imagine that they had already decided to move to a new apartment *before* they were confronted with the mistrust manipulation and the choice task. Hence, they might have evaluated the relevance of appliances choices in the context of the larger decision about moving to a new apartment, which was already made, which might have led to post-decisional dissonance reduction efforts in the face of negative information in the mistrust condition (Brehm, 1966). Third, a recent meta-analysis found that trust in institutions seems less critical for the uptake of private behaviors, such as reducing one's own electricity consumption, than for the uptake of public behaviors, such as the support of environmental policies and taxes (Cologna & Siegrist, 2020). The meta-analysis generally finds substantial variation in the influence of trust on pro-environmental behavior.

Furthermore, the lack of a clear moderation by the trust manipulation, and the unexpected negative interaction between the default manipulation and the trust manipulation in the U.S., which was significant when controlling for covariates (model 4, Table 1), might be due to a “ceiling effect” in this case. To illustrate, in the condition with pre-selected energy-efficient appliances, overall,² 84% of participants did not make a single switch within the list from an energy-efficient to a standard appliance. In the U.S. when headlines did not cast doubts on the real estate agent (e.g., trust condition), 91% of participants did not make a single switch from pre-selected energy-efficient to standard appliances, which dropped to 81% when headlines created mistrust (Fig. 3). This suggests that it would be very difficult for interventions to increase the number of energy-efficient choices further in the energy-efficient default condition – a ceiling effect. No constraints were present in the standard default condition. Perhaps this would have been different if the promoted behavior had been less convincing and less generally endorsed in society. In other words, in a scenario where default effects were expected to be weaker, or participants had stronger preferences against the promoted behavior, a “ceiling effect” might be absent, yielding more conclusive insights on the moderating role of trust. However, further studies should explore defaults and the role of trust in meaningful choices for participants.

In the U.S., but not in the Philippines, the trust manipulation had a positive main effect on energy-efficient choices, with participants choosing more energy-efficient appliances in the trust condition than in the mistrust condition. It is more surprising that the manipulation of trust in the choice architect did not affect choices in the Philippines than that it influenced choices in the U.S. The difference could be rooted in cultural values, such as perceived power distance – which defines the differences to which less powerful members of institutions within a country expect and accept that power is distributed unequally – which is substantially greater in the Philippines (94 out of 100) than in the U.S. (33 out of 100; Hofstede, 1991). Hence, people in the Philippines might be more inclined to follow a default, irrespective of their trust in the choice architect, whereas trust might be a stronger decisional cue in the U.S.

This study confirms previous research finding that environmental concern increases energy-efficient choices in non-WEIRD (Kuhn et al., 2021; Never et al., 2022) and WEIRD countries (Berman Caggiano et al., 2021). Participants with higher environmental concerns chose more energy-efficient appliances in both countries. However, this effect was stronger in the U.S. than in the Philippines. A possible reason could be the differences in familiarity with products labeled as energy-efficient. In the U.S., energy labeling was introduced for all major appliances and lighting in 1980. In the Philippines, labeling was introduced for some products in 1993. Hence, it might be that the concept of energy-efficiency and its environmental impacts are more established in the U.S. than in the Philippines, therefore associating the choice of

energy-efficient appliances more with environmental concern in the former than in the latter context.

At any rate, the study can inform applications of defaults. For example, the results speak for the efficacy of regulations such as the EU eco-design directive, which requires that the “default setting of the household dishwasher shall be the most energy-efficient automatic programme, unless the user selects a different programme” (Directive 2009/125/EC, 2012). Further, procurement regulations for government undertakings could benefit from these findings. For example, energy-efficient appliances could be made mandatory for procurements in public buildings. It has been suggested, that energy providers could offer green energy tariffs by default to new customers (Pichert & Katsikopoulos, 2008; Vetter & Kutzner, 2016). Equally, in new housing projects energy-efficient appliances could automatically be offered to customers before they move in. Further studies should explore trust as a moderator of defaults in such real-world settings.

7. Limitations

This study has the usual limitations of survey studies, including hypothetical choice tasks and subjects recruited from an online panel. We focus primarily on the mechanism of implied endorsement, which might require cognitive processing and involvement, which could be problematic in a hypothetical choice task. In short, people might not have cared as much about the choices as they would have in real life. Hence, it might be that trust is a stronger moderator in choice situations that succeed in creating greater involvement. To test this, future studies should try to increase involvement in the tasks further.

The study also downplays the attention to the price and its influence on decisions about electronic appliances. We told participants that the initial higher costs of energy-efficient appliances would be saved within three years. Still, we cannot preclude that price consideration influenced decisions. Market prices for electronic appliances and electricity, which vary between the U.S. and the Philippines, could also have affected decisions. Since the down-payment price is often viewed as a barrier for consumers to choose energy-efficient products, downplaying the price in this experiment might have inflated the default effect.

Further, our research design, comparing an opt-in (here: pre-selected standard appliances) to an opt-out (here: pre-selected energy efficient appliances) default condition, does not allow us to conclude the effectiveness of defaults compared to an active choice format (without a pre-selected option). Previous studies have found for simple dichotomous choice situations that, active-choice interventions can be equally effective as default interventions (Carroll et al., 2009; Keller et al., 2011; Toft et al., 2014). An active-choice format requires more cognitive engagement and is, therefore, more effortful and likely to trigger different evaluative processes than a default-choice format which steers cognitions towards the default in particular ways (e.g., Davidai et al., 2012). By virtue of triggering more deliberation, trust might actually be of particular relevance in an active choice format. Further research might explore the influence of trust on active choice in comparison to default interventions.

Studies comparing different country contexts can only consider a limited number of covariates that could potentially explain country differences. In this study, environmental concern was the only included covariate that contributed to predicting choices. Measures from the general preference survey (Falk et al., 2018) were not predictive of these choices. Still, there might be omitted variables that influence the studied decision processes.

All authors of this article are male and from WEIRD countries, which is a critical limitation of this study. We are increasingly becoming aware of the coloniality and harm that comes with excluding local researchers and their perspectives (G. Adams et al., 2018; Readsura Decolonial Editorial Collective, 2022). Our own epidemic distance from the site of data collection could increase the likelihood that we interpreted the results in ways that misrepresent local perspectives. We encourage other

² With both country datasets combined.

researchers to critically reflect on their epistemologies and foster collaboration with local researchers outside of WEIRD countries.

Furthermore, we agree with those critically reflecting on the role of behavioral research in fostering narratives of individual responsibility (see [Chater & Loewenstein, 2022](#)), potentially distracting from the necessity of societal change. We urge the field to explore avenues to motivate individual attitudes and behavior change alongside social and political transformation and to study the interplay between small-scale/individual activities and macro factors that could lead to a just ecological transformation (M. [Adams, 2021](#); [Wullenkord et al., 2021](#)).

8. Conclusions

This study provides new evidence supporting the robustness of default effects across WEIRD and non-WEIRD countries and in the face of mistrust in the choice architect. This is good news for those wanting to make use of default interventions. However, it is essential to acknowledge the ethical risks connected with the use of defaults. Defaults that are consciously set by a choice architect can be considered quasi-paternalistic interventions that bypass reflective thinking ([Hansen &](#)

[Jespersen, 2013](#)) – a means of manipulating people against their will and even against their best interest. The robustness of the default effect underscores how important it is that default interventions are rooted in democratic procedures – if possible, actively and disclosed.

Author note

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Declaration of competing interest

The authors whose names and affiliations are listed below have no affiliations with or involvement in any organization or entity with any financial interest, or non-financial interest in the subject matter or materials discussed in this manuscript. There are no conflict of interest regarding the manuscript with the title “No Trust in the Choice Architect? No Problem! On the Minor Role of Trust for the Effectiveness of Default Interventions Promoting the Choice of Energy-Efficient Appliances” submitted to the journal of Energy Research & Social Science.

Appendix 1

Intro Scenario

Now, imagine you consider renting a newly constructed apartment rented out by the Better Living Real Estate Group. Right now, this is your only available option. You have therefore decided to take it and signed the contract.

Better Living has a programme that supports your move to the new apartment and is asking you to fill out an online form in which you choose amenities for your new home. Before you fill out the form you come across some recent headlines from news websites that could be relevant in this situation.

Please read these headlines carefully and indicate how interesting you perceive each article to be for your situation. To make sure that you read them carefully, we will ask a simple question about the headlines after the task.

Appendix 2

Trust Manipulation

Domain	Positive Headlines		Negative Headlines	
	U.S.	Philippines	U.S.	Philippines
trust in choice architect	Award Winners: Affordable high-quality housing and community project by Better Living wins the 2021 American Residential Real Estate Award The Better Living Real Estate Group has been announced as the overall winner of the annual Residential Real Estate Award for its diverse building programme. The 2021 award winner has been praised for excellence in affordable housing design and dedication to its customer needs.	Award Winners: Affordable high-quality housing and community project by Better Living wins the 2021 Philippine Residential Real Estate Award The Better Living Real Estate Group has been announced as the overall winner of the annual Residential Real Estate Award for its diverse building programme. The 2021 award winner has been praised for excellence in affordable housing design and dedication to its customer needs.	Scammers indicted: Multimillion-dollar real-estate scheme by Better Living tied to fraudulent listings and sales The Federal Bureau of Investigation (FBI) and the Federal Deposit Insurance Corporation (FDIC) warn the public to be vigilant against fake agents engaged in illegal real estate transactions following the investigations against the Better Living Real Estate Group. The company faces federal charges alleging that they orchestrated several real estate fraud scams. In these scams, Better Living supposedly listed homes without the owners' consent and collected money from multiple would-be buyers for each of the not-for-sale homes.	Scammers indicted: Multimillion-dollar real-estate scheme by Better Living tied to fraudulent listings and sales The National Bureau of Investigation (NBI) together with the Department of Human Settlements and Urban Development (DHSUD) warn the public to be vigilant against fake agents engaged in illegal real estate transactions following the investigations against the Better Living Real Estate Group. The company faces federal charges alleging that they orchestrated several real estate fraud scams. In these scams, Better Living supposedly listed homes without the owners' consent and collected money from multiple would-be buyers for each of the not-for-sale homes.

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(continued)

Domain	Positive Headlines		Negative Headlines	
	U.S.	Philippines	U.S.	Philippines
trust in the law	More transparency in real estate: Memorandum passed to combat corruption in U.S. housing sector The recently issued anti-corruption memorandum by the current U.S. administration will ensure the integrity of services and establish new transparent structures for the housing sector.	More transparency in real estate: Housing department forms anti-corruption committee The Department of Human Settlements and Urban Development (DHSUD) has formed an Anti-Corruption Committee to ensure the integrity of services and establish new transparent structures for the housing sector.	Money laundering: Stream of foreign money flows through real estate Shell companies are used to shield the identities of buyers in the competition to secure condos. According to the Financial Crimes Enforcement Network (FinCEN), these companies elude the law by using real estate transactions to launder dirty money.	Money laundering: Stream of foreign money flows through real estate Shell companies are used to shield the identities of buyers in the competition to secure condos. According to the Anti-Money Laundering Council (AMLC), these companies elude the law by using real estate transactions to launder dirty money.
buyers trust	Buyers, Notice: Report shows real estate sector can be trusted According to a report by the U.S. Secret Service Global Investigation Operations Center (GIOCC), real estate is one of the most trustworthy business sectors in the U.S.	Buyers, Notice: Report shows real estate sector can be trusted According to a report by the National Bureau of Investigation, real estate is one of the most trustworthy business sectors in the Philippines.	Buyers, Alert: Report shows real estate sector not to be trusted According to a report by the U.S. Secret Service Global Investigation Operations Center (GIOCC), real estate is one of the least trustworthy business sectors in the U.S. Schemes to defraud owners of their homes are becoming more frequent.	Buyers, Alert: Report shows real estate sector not to be trusted According to a report by the National Bureau of Investigation (NBI), real estate is one of the least trustful business sectors in the Philippines. Schemes to defraud owners of their homes are becoming more frequent.
trust in the market	The rise of the housing market: What everyone gets wrong about housing in the U.S. Today, most public housing developments do, in fact, deliver on their promise of providing adequate, affordable housing to people in need, but they remain stigmatised because of the negative associations with distressed public housing.	The rise of the housing market: What everyone gets wrong about housing in the U.S. Today, most public housing developments do, in fact, deliver on their promise of providing adequate, affordable housing to people in need, but they remain stigmatised because of the negative associations with distressed public housing.	The decline of the housing market: U.S. housing market will soon be in free-fall Since the beginning of the pandemic, housing prices are surging to new records with no end in sight. They are being fuelled by historically low interest rates, high demand, and low supply, leading to a hyper-competitive market, where by now more than half of homes are sold above list price. The U.S. housing market is once again a bubble ready to burst.	The decline of the housing market: Philippine housing market will soon be in free-fall Since the beginning of the pandemic, housing prices are surging to new records with no end in sight. They are being fuelled by historically low interest rates, high demand, and low supply, leading to a hyper-competitive market, where by now more than half of homes are sold above list price. The Philippines' decade-long housing market boom is over, with a severe coronavirus-induced economic recession.
trust in construction	Americans deserve not just better, but the best: Administration is seeking largest ever investment in building resilient, quality houses In their social spending bill, called Build Back Better, the current administration wants to make one of the biggest investments in history and provide low-cost housing projects for all Americans.	Filipinos deserve not just better, but the best: Administration is seeking largest ever investment in building resilient, quality houses The National Housing Authority (NHA) together with the Home Development Mutual Fund, Social Housing Finance Corporation, and the National Home Mortgage Finance Corporation want to make the biggest investment in history and provide low-cost housing projects for all Filipinos.	Poor construction: Decline in quality within the building industry There has been a noticeable decline in the quality of construction projects. For example, according to a study by the Department of Earth and Environment of the Florida International University, the catastrophic collapse of a Florida high-rise was predictable and could have been prevented. The construction plan was a complete failure and the building has been sinking at an alarming rate since the last 10 years due to low-quality material and poor planning.	Poor construction: Decline in quality within the building industry There has been a noticeable decline in the quality of construction projects. For example, two subcontractors recently testified before the Senate and claimed irregularities in the construction of the Bagong Pag-asa Housing Project by the National Housing Authority (NHA). The property development plan is a complete failure and high-quality building material is not being delivered.

Appendix 3

Attention Check

The following is an attention check, to test whether you read the previous headlines attentively. If you fail to answer this question correctly, the study will terminate early.

What was the news about Better Living?

- o It won an awards for affordable high-quality housing (*correct answer in trust condition*)
- o It faces federal charges linked to a frau scam (*correct answer in mistrust condition*)
- o It plans to build the tallest construction worldwide
- o It merges with a different company becoming one of the largest real estate companies

Appendix 4

Manipulation Check

To what extent do you trust that Better Living has the customers' best interests in mind?

Please indicate your answer on a scale from 0 to 10, where 0 means "do not trust at all" and a 10 means "completely trust".

To what extent do you trust that real estate companies in general have the customers' best interests in mind?

Please indicate your answer on a scale from 0 to 10, where 0 means "do not trust at all" and a 10 means "completely trust".

Appendix 5

Appliances Choice Task (based on Steffell et al., 2016)

Opt-in Default



Better Living Real Estate Group

Dear tenant,

We are glad that you chose to rent with us. We want to make moving in to your new apartment as easy and satisfying as possible for you.

We have therefore preinstalled the following amenities in your new home. They are standard amenities, but if you want we can replace them with energy-efficient alternatives. Compared to the standard amenities, the energy-efficient amenities cost more, but due to the savings on the electricity bill, the operating costs are lower. We estimate that the energy-efficient amenities save the equivalent of the initial higher costs in about three years.

If you want to keep the standard amenities, you do not need to do anything. If you want us to replace one or more with energy-efficient alternatives, you can indicate so by changing the tick-boxes. Submit your choices by continuing to the next page.

	standard	energy-efficient
freezer	☒	☐
refrigerator	☒	☐
washing machine	☒	☐
electric oven and stove	☒	☐
air conditioner	☒	☐

Opt-out Default



Better Living Real Estate Group

Dear tenant,

We are glad that you chose to rent with us. We want to make moving in to your new apartment as easy and satisfying as possible for you.

We have therefore preinstalled the following amenities in your new home. They are energy-efficient amenities, but if you want we can replace them with standard alternatives. Compared to the standard amenities, the energy-efficient amenities cost more, but due to the savings on the electricity bill, the operating costs are lower. We estimate that the energy-efficient amenities save the equivalent of the initial higher costs in about three years.

If you want to keep the energy-efficient amenities, you do not need to do anything. If you want us to replace one or more with standard alternatives, you can indicate so by changing the tick-boxes. Submit your choices by continuing to the next page.

	energy-efficient	standard
refrigerator	☒	☐
electric oven and stove	☒	☐
air conditioner	☒	☐
freezer	☒	☐
washing machine	☒	☐

Appendix 6

Freedom to Choose (based on Dillard & Shen, 2005; Shen & Dillard, 2005)

Note: Freedom to Choose was measured on a 7 point scale with the scale of completely disagree, strongly disagree, disagree, neither disagree nor agree, agree, strongly agree, completely agree.

How much do you agree or disagree with the following statements regarding when you were asked to select your preferred amenities?

I was free to decide for myself.

An attempt was made to make a decision for me.

Attempts were made to manipulate me.

I was put under pressure.

Appendix 7

Environmental Concern (based on Thøgersen et al., 2010)

Note: Environmental Concern was measured on a 7 point scale with the scale of completely disagree, strongly disagree, disagree, neither disagree nor agree, agree, strongly agree, completely agree.

How much do you agree or disagree with the following statements?

It is important to me that the products that I use do not harm the environment.

I consider the potential environmental impact of my actions when making many of my decisions.

My purchase habits are affected by my concern for our environment.

I am concerned about wasting the resources of our planet.

I would describe myself as environmentally responsible.

I am willing to restrict myself in order to take actions that are more environmentally friendly.

Appendix 8

Global Preferences (based on Falk et al., 2018)

In general, how willing are you to give up something that is beneficial for you today in order to benefit more from that in the future?

Please indicate your answer on a scale from 0 to 10, where 0 means you are “completely unwilling to do so” and a 10 means you are “very willing to do so”.

In general, how willing or unwilling are you to take risks?

Please use a scale from 0 to 10, where 0 means “completely unwilling to take risks” and a 10 means you are “very willing to take risks”.

In general, how much do you trust that people have only the best intentions in mind?

Please indicate your answer on a scale from 0 to 10, where 0 means “do not trust at all” and a 10 means “completely trust”.

Appendix 9

Table 1
Hierarchical regression analysis predicting the number of energy-efficient appliances chosen in the Philippines ($N = 909$) and in the U.S. ($N = 945$)

Constant	Philippines					U.S.				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
	0.00 \ 3.49*** (3.38, 3.60)	0.00 \ 3.49*** (3.38, 3.60)	0.00 \ 3.49*** (3.38, 3.60)	0.00 \ 3.49*** (3.38, 3.60)	0.00 \ 3.43*** (3.31, 3.56)	0.00 \ 3.58*** (3.47, 3.69)	0.00 \ 3.57*** (3.46, 3.68)	0.00 \ 3.57*** (3.46, 3.68)	0.00 \ 3.57*** (3.47, 3.68)	0.00 \ 3.69*** (3.09, 4.28)
<i>Independent variables:</i>										
Default Condition (−0.5 = standard, 0.5 = efficient)	0.52 \ 2.08*** (1.86, 2.31)	0.52 \ 2.08*** (1.86, 2.31)	0.52 \ 2.09*** (1.86, 2.31)	0.52 \ 2.10*** (1.88, 2.32)	0.52 \ 2.09*** (1.87, 2.31)	0.51 \ 2.08*** (1.85, 2.30)	0.52 \ 2.09*** (1.87, 2.31)	0.52 \ 2.09*** (1.87, 2.31)	0.52 \ 2.11*** (1.90, 2.32)	0.51 \ 2.08*** (1.87, 2.29)
TrustCondition (−0.5 = Mistrust, 0.5 = Trust)		−0.01 \ −0.04 (−0.27, 0.18)	−0.01 \ −0.04 (−0.27, 0.18)	−0.01 \ −0.06 (−0.28, 0.16)	−0.02 \ −0.07 (−0.29, 0.15)		0.14 \ 0.57*** (0.35, 0.79)	0.14 \ 0.57*** (0.35, 0.79)	0.13 \ 0.51*** (0.30, 0.72)	0.13 \ 0.51*** (0.30, 0.72)
Default X Trust Condition			−0.01 \ −0.06 (−0.50, 0.39)	−0.00 \ −0.04 (−0.48, 0.41)	−0.01 \ −0.05 (−0.49, 0.40)			−0.05 \ −0.41 (−0.85, 0.03)	−0.06 \ −0.50* (−0.93, −0.08)	−0.06 \ −0.51* (−0.93, −0.09)
<i>Covariates:</i>										
Environmental Concern				0.07 \ 0.14* (0.02, 0.27)	0.06 \ 0.13* (0.01, 0.26)				0.23 \ 0.40*** (0.31, 0.50)	0.23 \ 0.40*** (0.31, 0.50)
Patience				0.02 \ 0.03 (−0.03, 0.10)	0.03 \ 0.04 (−0.03, 0.11)				−0.04 \ −0.04 (−0.10, 0.02)	−0.04 \ −0.03 (−0.09, 0.03)
Risk				0.02 \ 0.04 (−0.03, 0.11)	0.03 \ 0.04 (−0.04, 0.11)				−0.03 \ −0.02 (−0.08, 0.03)	−0.00 \ −0.00 (−0.05, 0.06)
General Trust				−0.00 \ −0.00 (−0.06, 0.06)	0.00 \ 0.00 (−0.06, 0.06)				−0.02 \ −0.02 (−0.07, 0.03)	−0.04 \ −0.03 (−0.08, 0.02)
Age					0.05 \ 0.01 (−0.00, 0.02)					0.12 \ 0.02*** (0.01, 0.02)
Gender (−0.5 = Male, 0 = Nonbinary, 0.5 = Female)					0.05 \ 0.19 (−0.04, 0.42)					0.04 \ 0.18 (−0.05, 0.42)
Income					0.01 \ 0.01 (−0.06, 0.08)					0.06 \ 0.09* (0.02, 0.17)
Location (−0.5 = Suburban, 0.5 = Urban)					0.04 \ 0.18 <(−0.06, 0.43)					−0.01 \ −0.06 (−0.32, 0.20)
Adjusted R ²	0.27	0.27	0.27	0.28	0.28	0.26	0.28	0.28	0.33	0.34

Note: Standardized coefficients are shown before and unstandardized coefficients after the backslash. Confidence intervals of the unstandardized coefficients are in parenthesis. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

Appendix 10

Table 2
Hierarchical regression analysis predicting the number of energy-efficient appliances chosen with the two country datasets combined ($N = 1854$).

Constant	(1)	(2)	(3)	(4)	(5)	(6)
	−0.00 \ 3.53*** (3.45, 3.61)	0.00 \ 3.53*** (3.45, 3.61)	−0.00 \ 3.53*** (3.45, 3.61)	−0.00 \ 3.52*** (3.43, 3.60)	−0.00 \ 3.52*** (3.44, 3.59)	−0.04 \ 3.52*** (3.31, 3.58)
<i>Independent variables:</i>						
Country (−0.5 = Philippines, 0.5 = U.S.)	0.02 \ 0.09 (−0.07, 0.24)	0.02 \ 0.08 (−0.07, 0.24)	0.02 \ 0.08 (−0.07, 0.24)	0.07 \ 0.29** (0.12, 0.47)	−0.01 \ −0.06 (−0.31, 0.19)	−0.00 \ 0.01 (−0.28, 0.25)
Default Condition (−0.5 = standard, 0.5 = efficient)	0.52 \ 2.08*** (1.92, 2.24)	0.52 \ 2.09*** (1.93, 2.24)	0.52 \ 2.09*** (1.93, 2.24)	0.52 \ 2.10*** (1.95, 2.26)	0.52 \ 2.09*** (1.94, 2.24)	0.52 \ 2.08*** (1.93, 2.24)
Default Condition X Country	−0.00 \ −0.01 (−0.32, 0.30)	0.00 \ 0.01 (−0.31, 0.32)	0.00 \ 0.01 (−0.30, 0.32)	0.00 \ 0.01 (−0.30, 0.31)	−0.00 \ −0.02 (−0.32, 0.29)	−0.00 \ −0.01 (−0.32, 0.29)
Trust Condition (−0.5 = Mistrust, 0.5 = Trust)		0.07 \ 0.26*** (0.11, 0.42)	0.07 \ 0.26** (0.11, 0.42)	0.05 \ 0.21** (0.06, 0.37)	0.05 \ 0.21** (0.05, 0.36)	0.06 \ 0.22** (0.07, 0.37)
Trust Condition X Country		0.08 \ 0.62*** (0.30, 0.93)	0.08 \ 0.61*** (0.30, 0.92)	0.08 \ 0.60*** (0.30, 0.91)	0.08 \ 0.62*** (0.32, 0.93)	0.07 \ 0.58*** (0.28, 0.89)
Default X Trust Condition			−0.03 \ −0.23 (−0.54, 0.08)	−0.03 \ −0.25 (−0.56, 0.05)	−0.03 \ −0.26 (−0.56, 0.05)	−0.03 \ −0.28 (−0.58, 0.03)
Default X Trust Condition X Country			−0.02 \ −0.35 (−0.98, 0.27)	−0.03 \ −0.48 (−1.10, 0.13)	−0.03 \ −0.48 (−1.09, 0.13)	−0.03 \ −0.46 (−1.07, 0.15)
<i>Covariates:</i>						
Environmental Concern				0.17 \ 0.31*** (0.23, 0.38)	0.17 \ 0.30*** (0.22, 0.37)	0.15 \ 0.27*** (0.19, 0.35)
Patience				−0.01 \ −0.01 (−0.05, 0.04)	−0.00 \ −0.00 (−0.05, 0.04)	−0.00 \ −0.00 (−0.04, 0.05)
Risk				−0.00 \ −0.00 (−0.05, 0.04)	0.01 \ 0.01 (−0.03, 0.05)	0.02 \ 0.02 (−0.03, 0.06)
General Trust				−0.02 \ −0.01 (−0.05, 0.02)	−0.02 \ −0.02 (−0.06, 0.02)	−0.02 \ −0.02 (−0.05, 0.03)
Age					0.14 \ 0.01*** (0.01, 0.02)	0.12 \ 0.01*** (0.01, 0.02)
Gender (−0.5 = Male, 0 = Nonbinary, 0.5 = Female)					0.05 \ 0.19* (0.03, 0.35)	0.05 \ 0.19* (0.02, 0.35)
Income						
Location (−0.5 = Suburban, 0.5 = Urban)					0.04 \ 0.05 (−0.00, 0.10)	0.04 \ 0.05* (0.00, 0.10)
E. Concern X Country					0.02 \ 0.07 (−0.10, 0.25)	0.01 \ 0.06 (−0.12, 0.24)
Patience X Country						0.08 \ 0.27*** (0.12, 0.43)
Risk X Country						−0.03 \ −0.07 (−0.16, 0.03)
General Trust X Country						−0.02 \ −0.03 (−0.12, 0.06)
Age X Country						−0.02 \ −0.04 (−0.11, 0.04)
Gender X Country						0.03 \ 0.01 (0.01, 0.02)
Income X Country						−0.00 \ −0.01 (−0.34, 0.32)
Area X Country						0.03 \ 0.08 (−0.02, 0.18)
						−0.03 \ −0.24 (−0.60, 0.12)
Adjusted R ²	0.27	0.28	0.28	0.30	0.31	0.31

Note: Standardized coefficients are shown before and unstandardized coefficients after the backslash. Confidence intervals of the unstandardized coefficients are in parenthesis. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

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